

PROFESSIONAL SUMMARY

Founding engineer at **Suga** and team lead at **Illumibot** with a bias for ownership. Solo-shipped a production system handling **1M+ network requests** at launch and generating **\$30K in new client revenue**.
Led cross-functional delivery across a **20+ person engineering team**, managing **5+ parallel projects** spanning code review, architecture decisions, and end-to-end ownership.
Stack: **TypeScript, React, Next.js, Python, Flutter, Dart**.

WORK EXPERIENCE

- **Software Development Engineer & Team Lead - Arroyodev LLC**08/2025 - Present

Full-stack engineering for **Illumibot.ai** (10,000+ users), a personalized projection-mapping ecosystem within a **multi-repo microservices architecture**. Promoted to Team Lead on select engagements and **Joint Co-Lead** on the core Illumibot app after consistent end-to-end ownership.

- **Flagship Project - Transworld Floor Plan (HAA 2026)**: Shipped solo from blank repo to production under hard deadline. Peaked at **1M+ network requests** at launch with zero downtime; earned **500+ positive customer reports**; generated **\$30,000** in new ad/sponsorship revenue for the client by transforming a static PDF into a live interactive exhibit map.
 - **Mobile & Algorithmic Engineering**: Maintaining the core **Flutter** application; optimizing spatial projection algorithms and computer-vision feature-point detection to improve mapping accuracy by **25%**.
 - **Full-Stack Web Development**: Refining the **Next.js** e-commerce and management portal with responsive design patterns and payment/user-management API integration.
 - **Backend & Data Infrastructure**: Authoring **Node.js/Python** automation against **Firebase** to resolve data conflicts, manage NoSQL schema migrations, and keep data synchronized across multi-user sessions.
- **Founding Engineer - Suga**10/2025 - Present

AI-powered workflow operating system for customer support. Record an agent solving a ticket once → AI synthesizes an executable workflow → human approves → runs automatically across 11 platform integrations. Zero-config - no knowledge base, no rules engine, no developer required.

- **Session-to-Workflow Pipeline**: **Chrome extension** captures agent actions; **Google Gemini API** analyzer synthesizes a structured executable workflow; **Next.js** visual builder for human review/edit; multi-tenant **Fastify + Supabase (PostgreSQL)** execution engine deploys across 11 platform connectors.
 - **Human-in-the-Loop Approval** (Pending → Approved → Deployed) with workflow versioning + rollback - eliminates AI-hallucination risk while preserving zero-config UX.
 - **Real-Time Infrastructure**: Unified inbox with **Socket.IO** chat, **LiveKit + Twilio SIP + Deepgram STT** voice, **SendGrid + Gmail/Outlook/IMAP** email; **Stripe** billing; **Sentry** observability. **Placed 2nd of 30+ teams at USM-VOXO Hackathon (\$2,500)**.

PROJECTS & RESEARCH

- **Roadmap.io - Personalized Job-Matching Platform** [link]03/2025 - Present

- Built a responsive **React** frontend with **Tailwind CSS** via a self-created design system for consistent UX across breakpoints.
 - Rebuilt the mobile **Sigma.js** graph view with touch controls and adaptive sizing for improved mobile engagement.
 - Refactored monolithic CSS to utility-first, **reducing bundle size by 60%** and **cutting page-load from 3.2s to 1.8s** via lazy loading.
- **Robust and Encrypted Federated Learning (1st Place, USM Symposium - Cybersecurity)** [link]08/2024 - 08/2025

- **Improved accuracy retention by 12%** over baseline encrypted aggregation by orchestrating a 14-configuration federated learning pipeline with **Python, Flower Framework**, and **PyTorch**.
 - Shipped a real-time **Matplotlib** dashboard visualizing learning curves and cost-per-accuracy tradeoffs across configurations.
 - Engineered a normalized-cost-metric visualization tool, enabling stakeholders to compare system configurations **5× faster** than previous manual methodologies.

TECHNICAL SKILLS

- **Languages**: TypeScript, JavaScript, Python, SQL, Dart, C, C++, C#
- **Frontend & Mobile**: React, Next.js, Tailwind CSS, Flutter
- **Backend & Data**: Node.js, Express.js, Fastify, REST APIs, WebSocket, Socket.IO, PostgreSQL, Supabase, Firebase
- **AI, Cloud & Tools**: PyTorch, Flower (Federated Learning), CKKS Homomorphic Encryption, Google Gemini API, Docker, Vercel, Git, CI/CD (GitHub Actions), Stripe, LiveKit + Twilio SIP + Deepgram STT, SendGrid, Gmail/Outlook/IMAP, Sentry

EDUCATION

- **University of Southern Mississippi**01/2023 - 05/2026

*Bachelor of Science in Computer Science (Honors), Minor in Economic Data Analysis***GPA: 4.00, President's List**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering, Machine Learning, Discrete Mathematics

Awards & Honors
 - **USM Undergraduate Research Symposium (CS, Cybersecurity Track) - 1st Place**: Oct 2025 - 1st of 50+ researchers for paper "Robust and Encrypted Federated Learning Under Constraint."
 - **Full-Tuition and Housing Scholarship**: \$60,000 (Jan 2023) - Awarded for academic excellence and leadership.
 - **Honors Keystone Award**: \$2,000 (Aug 2024) - Recognized for academic and research excellence.
 - **Eagle SPUR Grant**: \$1,500 (May 2025) - Fellowship for secure software-engineering research.
 - **USM-VOXO Hackathon - 2nd Place**: \$2,500 (Oct 2025) - Ranked 2nd of 30+ teams for rapid prototyping and execution (Suga).